

Bubble Reactor Self Study Project

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(a) Introduction to the Reactor

What Is a Bubble Reactor?

Bubble reactor is a type of gas liquid reactor typically modeled in a circular column where liquid is poured to fill the column co or counter current direction. A sparge of gas enters the column ,typically from the bottom, creating bubbles. Inside the column, the gas superficial velocity is greater than liquids velocity making the reactor Gas phase dominating. Some Bubble reactors can even include three phases inside their column, in this case the solid phase are catalysts.

Bubble reactors can be analyzed by the different flow regimes created by the bubble which in fact have a big importance on the modeling of the system. At low Gas velocity we get a Bubbly flow regime that creates a small bubble with little to no bubble coalescence. At high Gas velocity with small diameter the bubbles will interact with each other to create a bubble coalescence as large as the diameter of the column called Slug flow regime, and with increased diameter of the column there will be Churn turbulent regime where there are multiple bubble size distributions. In many cases Bubbly flow regime is favored since it is the easiest to model and is used in industry. The other models are seen more in experimental laboratories. There are also varying types of columns like multistage, multi-channel, Jet reactor etc.

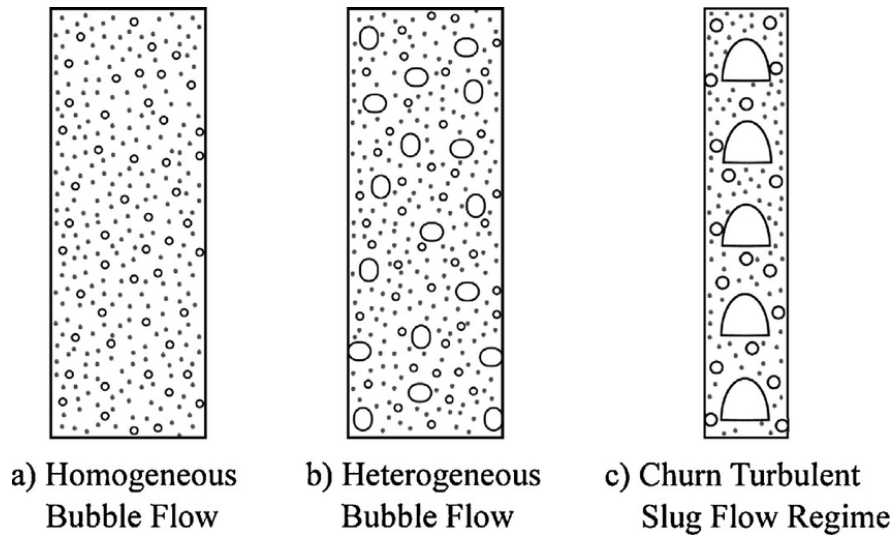


Figure 1. Flow regimes

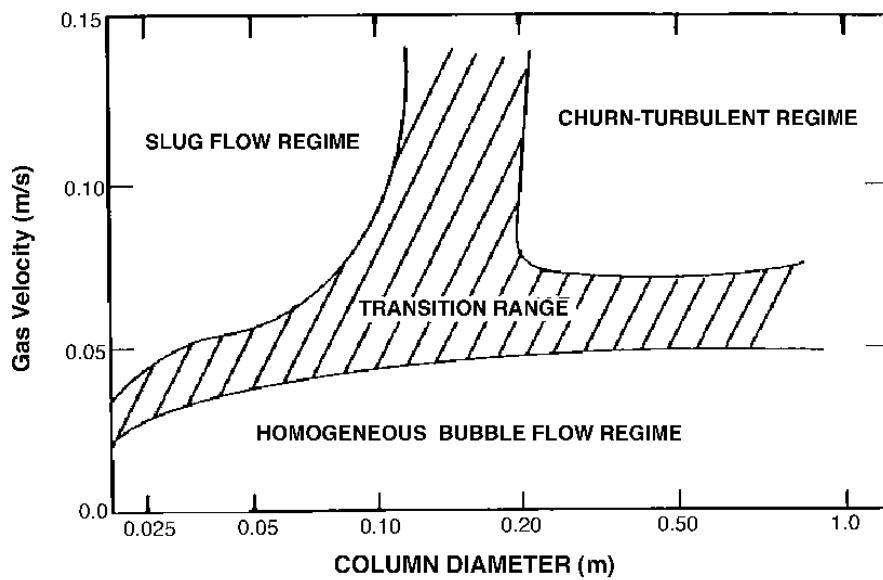


Figure 2.

The design of the reactor should consider these various parameters. Maximization of surface area. Maximization of Mass Transfer Coefficient by reducing bubble diameter. Gas holdup which is crucial for gas phase residence time and mass transfer between liquid and gas. The bubble shape, size, and velocity should always be measured and known to be able to design the reactor. The Bubble itself also depends on many forces like buoyancy, flux of gas, surface tension, and drag force.

Some advantages of using the Bubble reactor from others is that it is very low cost and low maintenance because there are not many moving parts to the reactor itself. Temperature can also be controlled easily for safety reasons. Most importantly the reactor can have a three-phase column where other reactors are only one phase. Some disadvantages can be a High pressure drop of the gas and having to figure out very specific gas/liquid ratios so as not to create bubble coalescence.

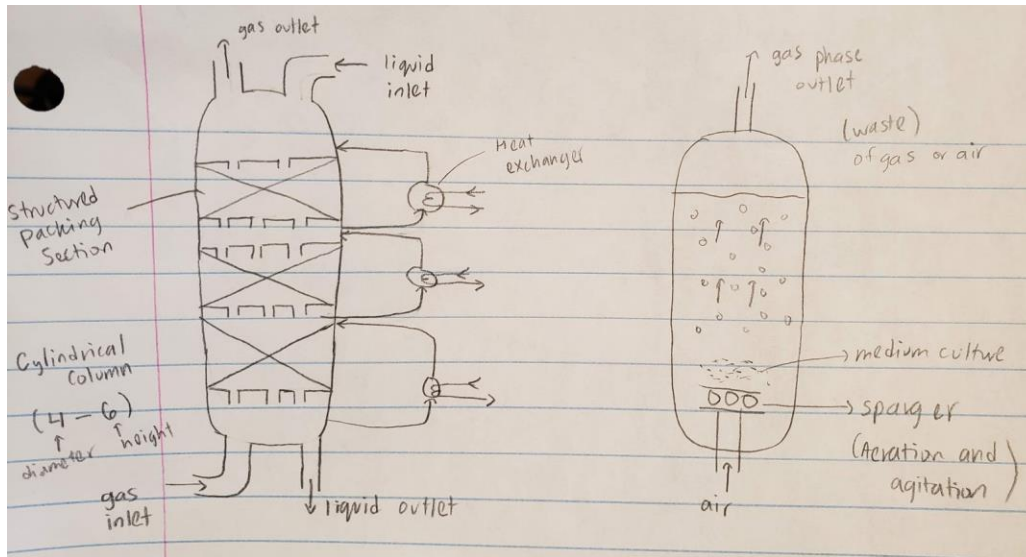
(b)Detailed Process and its Importance

Bubble column reactors are heavily used in chemical, biochemical, petrochemical, and metallurgical industries. The reactor can be used in chemical processes with reactions such as oxidation, hydrogenation, polymerization, creating synthetic fuels (gas conversion process), fermentation and wastewater treatment (biochemical processes).

In industrial settings, bubble columns usually operate based on a length-to-diameter ratio (aspect ratio) of at least 5. In the biochemical industry, the range can vary between 2 to 5.

There are two different modes of bubble column reactors: continuous and semi-batch. In continuous operation, the inlet gas and the suspension flow upward in the column and the suspension that exits the column is recycled back to the feed tank. There is also liquid flowing at a superficial velocity a magnitude lower than the gas inlet velocity. In the semi-batch operation, only gas is bubbled upward as suspension remains stationary. Spargers are used to generate uniform size bubbles and allow for the bubbles to spread uniformly.

In general, there is a gas inlet from the bottom of the column which is sparged (through a porous plate) to provide aeration and agitation for the suspension/media/culture which can remain in the column (semi-batch operation) or be recycled into the feed tank (continuous operation). Heat exchanger(s) can be added to the bubble column to ensure a constant temperature.



(c) Design Equations

Mole Balance:

$$\underbrace{\epsilon_{\text{bed}} \cdot \frac{\partial c_G}{\partial t}}_{\text{Accumulation}} = \underbrace{-\frac{\partial}{\partial z} (u_G \cdot c_G)}_{\text{Advection}} + \underbrace{\sum_i v_i \cdot \eta_{\text{cat}} \cdot (1 - \epsilon_{\text{bed}}) \cdot \rho_S \cdot r_{2\text{PM}}}_{\text{Reaction}}$$

Component Mole Balance:

$$\underbrace{\epsilon_{\text{bed}} \cdot \frac{\partial c_{i,G}}{\partial t}}_{\text{Accumulation}} = \underbrace{-\frac{\partial}{\partial z} (u_G \cdot c_{i,G})}_{\text{Advection}} + \underbrace{v_i \cdot \eta_{\text{cat}} \cdot (1 - \epsilon_{\text{bed}}) \cdot \rho_S \cdot r_{2\text{PM}}}_{\text{Reaction}}$$

Energy Balance:

$$\underbrace{(\rho_G \cdot c_{p,G} \cdot \epsilon_{\text{bed}} + \rho_S \cdot c_{p,S} \cdot (1 - \epsilon_{\text{bed}})) \cdot \frac{\partial T}{\partial t}}_{\text{Accumulation}} = \underbrace{-\frac{\partial}{\partial z} (\rho_G \cdot c_{p,G} \cdot u_G \cdot T)}_{\text{Advection}}$$

$$\underbrace{+ \eta_{\text{cat}} \cdot (1 - \epsilon_{\text{bed}}) \cdot \rho_S \cdot r_{2\text{PM}} \cdot (-\Delta h_r)}_{\text{Reaction heat}} - \underbrace{\frac{4 \cdot \alpha_{\text{eff}}}{d_{\text{tube}}} \cdot (T - T_{\text{cool}})}_{\text{Cooling}}$$

Pressure Drop [Ergun!]

$$\frac{\partial p}{\partial z} = -\frac{u_G}{d_p} \cdot \frac{1 - \epsilon_{\text{bed}}}{\epsilon_{\text{bed}}^3} \cdot \left(1.75 \cdot \rho_G \cdot u_G + 150 \cdot \frac{\mu_G}{d_p} \cdot (1 - \epsilon_{\text{bed}}) \right)$$

Designing and scaling up bubble columns

I) Hatta number: compares the rate of reaction in a liquid film to the rate of diffusion through the film.

For a reaction mth order in A and nth order in B

$$\text{Ha} = 2m+1 \cdot k_{m,n} \cdot C_{A,\text{im}}^{-1} C_{B,\text{bulk}} m D_A k_L$$

$$\text{Ha} < 0.3$$

Correlations~

NOTE: Multiple available depending on the scenario. Will only provide the simplest case.

1. Gas Holdup:

- a. Dimensionless key parameter for design purposes that characterizes the transport phenomena of bubble column systems which is basically defined as the volume fraction of gas phase occupied by the gas bubbles.

ex:

$$g = v_g 0.3 + 2V_g$$

2. Bubble Size:

a. Sauter diameter (dbS): Mean bubble diameter, calculated from the volume to surface

$$d_{bs} = 2 \epsilon M^{0.4} (L)^{0.6} G^{0.5} (GL)^{0.25}$$

3. Mass Transfer Coefficient

$$k_1 a = 0.476 V_g^{0.82}$$

$$a = \frac{6 \epsilon_g}{d_s}$$

4. Heat transfer coefficients

There are different coefficients depending on the type of column.

$$St = 0.1 (\text{Re } Fr \text{ Pr}^2)^{-0.25}$$

$$St = \frac{h_w}{\rho_l C_p V_g}$$

$$\text{Re} = \frac{V_g d_p \rho_l}{\mu_l}$$

$$Fr = \frac{V_g^2}{g D_c}$$

$$\text{Pr} = \frac{C_p \mu_l}{k_l}$$

(d) Example Solution

From the above link, I found an example that relates to the bubble reactor to work on.

Question:

Liquid water at 25°C is to be aerated in a bubble column where finely air bubbles with diameter d_B of 0.5 mm are injected cocurrently with the liquid. The interfacial contact area, a , between air and water can be calculated from the expression $a = 6\varepsilon/d_B$, where ε is the volume fraction of the injected air. The bubble column is 1.8 m high with a superficial liquid velocity of 0.2 m/s. The oxygen concentration of the inlet water is 0.12×10^{-4} kmol/m³

. The

saturated oxygen concentration is 2.67×10^{-4} kmol/m³

. Determine the oxygen concentration of

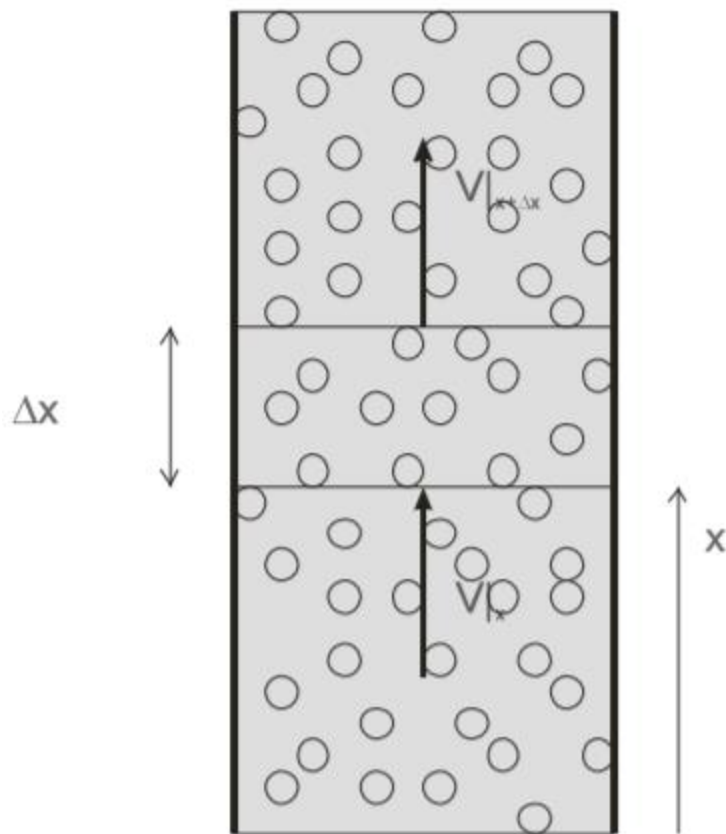
the outlet water if the mass transfer coefficient for the transfer of oxygen from the liquid

interface to the bulk water is 5.8×10^{-6} m/s. The diffusivity of oxygen in water is 2.42×10^{-9}

m

2

/s. The volume fraction of the injected air is 0.2.



Solution:

Step 1: Assume Steady State and do material balance.

$$C_A * U * A_c|_x - C_A * U * A_c|_{x+\Delta x} + (1 - \epsilon) * A_c * N_A|_x - (1 - \epsilon) * A_c * N_A|_{x+\Delta x} + k_c * (C_{sat} - C) * a * A_c * \Delta x = 0$$

Parameters to define:

U : superficial liquid velocity

A_c : cross sectional area of bubble column

ϵ : void fraction

C_A : concentration of oxygen in water

Csat: concentration of saturated oxygen in liquid

NA: molar flux of oxygen

Kc: the mass transfer coefficient

Step 2: Divide everything by Δx and take a limit that Δx goes to 0

$$-(1-\epsilon) \frac{dN_A}{dx} - U \frac{dC_A}{dx} + k_c (C_{sat} - C) a = 0$$

Step 3: Substitute $N_A = -D_{AB} \frac{dC_A}{dx}$ into the step 2 equation

$$(1-\epsilon) D_{AB} \frac{d^2 C_A}{dx^2} - U \frac{dC_A}{dx} + k_c (C_{sat} - C) a = 0$$

Step 4: Assume axial diffusion is negligible

$$D_{AB} \frac{d^2 C_A}{dx^2} = 0$$

$$-U \frac{dC_A}{dx} + k_c (C_{sat} - C) a = 0$$

Step 5: Separation of variables

$$\int_{C_0}^{C_L} \frac{dC_A}{(C_{sat} - C)} = \int_0^L \frac{dx}{U}$$

Step 6: Take the integrals on both sides

$$-\ln\left(\frac{C_{sat} - C_L}{C_{sat} - C_0}\right) = \frac{a k_c L}{U}$$

$$\rightarrow C_L = C_{sat} - (C_{sat} - C_0) \cdot e^{(-a k_c L / U)}$$

From the problem state, we know that $a = 6\epsilon/dB$

$$a k_c L / U = 6 \cdot \epsilon \cdot k_c \cdot L / (dB \cdot U) = 6 \cdot 0.2 \cdot 5.8E-6 \cdot 1.8 / (0.5E-3 \cdot 0.2) = 0.1253$$

Step 7: plug back into step 6 equation:

$$C_L = C_{\text{sat}} - (C_{\text{sat}} - C_0) * e^{(-a * k_c * L / U)} = 2.67E-4 - (2.67 - 0.12)E-4 * e^{(-0.1253)} = 0.42E-4 \text{ kmol/m}^3$$

(e) Safety Considerations

Since that Bubble Column Reaction (BCR) is mostly used in 2 phase reaction (liquid and gas), eruption and splashing may occur. Usually, for BCR, if the ratio of the height of the reactor to the diameter of the reactor exceeds 12, bubble coalescence will occur significantly because the center of the column has less resistance than the wall, the bubble will gradually concentrate toward to the center, so the bubble will combine together to form bigger bubbles, which might cause splashing as the large bubble exit the reactor column. Therefore, the size of the column would be an extremely important design factor if the liquid inside is corrosive or toxic. If the reactor is holding an exothermic reaction, pressure would be more difficult to control due to the heat release and temperature change. For exothermic reaction undergoing inside the BCR, the heat release from the reaction will be heating up the reactant liquid to its boiling point, which may extra gaseous substance other than the supplying gas, which may cause the total pressure inside the will be constantly increasing if the reactor is only designed to held certain pressure releasing. As the total pressure inside increases, once it reaches the reactor limit, an eruption occurs. Therefore, we should have considered the properties of the metal we used to build the reactor that can easily hold some temperature changes without having plastic deformation. We should also consider having a coolant stream covering the BCR reaction column to remove the heat generation. Overall, Bubble Column Reactor has a very simple construction, temperature and pressure would be relatively easy to control.

Appendix

Sources: [HyperLink with quick description for future reference]

<https://onlinelibrary.wiley.com/doi/pdf/10.1002/cjce.20327>

<https://www.slideshare.net/AkshithKanchi/bubble-column-reactor>

[http://web.ist.utl.pt/~ist11061/de/Equipamento/BubbleColumnReactors\(review\).pdf](http://web.ist.utl.pt/~ist11061/de/Equipamento/BubbleColumnReactors(review).pdf)

<https://reader.elsevier.com/reader/sd/pii/S0960148119316830?token=742BBD1EC5BC4B267B17ED137354F92CEA835DE64D3B5C80BF8FD40C4AEE91B7C02616866FFAB1185EAB95773EC3AA74>

<https://slideplayer.com/slide/4184752/>

[http://web.ist.utl.pt/~ist11061/de/Equipamento/BubbleColumnReactors\(review\).pdf](http://web.ist.utl.pt/~ist11061/de/Equipamento/BubbleColumnReactors(review).pdf)

<https://www.youtube.com/watch?v=z4LS4Z6U8hg>

<https://www.cpp.edu/~tknguyen/che313/pdf/chap3-1e.pdf>

Modeling of the transient behavior of a slurry bubble column reactor for CO₂ methanation, and comparison with a tube bundle reactor

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Design of a bubble column reactor

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